Forecasting Demand for National Research Supercomputing

Australian Academy of Science Internship Project

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Chapter 1

Forecasting Demand for National Research Supercomputing

Australian Academy of Science Internship Project

Chapter 2

Introduction

2.1 Project Background

High-performance computing (HPC) is an important part of Australia's research infrastructure, supporting computationally intensive research across a wide range of scientific fields.

Australia has two national Tier-1 HPC facilities accessible to researchers: the National Computational Infrastructure (NCI) and the Pawsey Supercomputing Research Centre. Several Tier-2 HPC facilities also operate across Australian research institutions.

As research becomes increasingly data- and compute-intensive, understanding future demand for HPC resources is important for planning Australia's national research infrastructure.

2.2 Project Purpose

The purpose of this project is to develop a robust forecast of Australian research demand for high-performance computing to 2030.

The forecast is intended to support national policy discussions around public compute investment and future research infrastructure planning.

The project combines data engineering, descriptive analysis and forecasting using historical Australian HPC data and other indicators that may help explain changes in research compute demand.

2.3 The Forecasting Problem

One of the main challenges is that observed HPC allocation does not necessarily represent actual research demand.

National allocation schemes operate under limited available capacity. As a result, allocated compute represents the amount of resource awarded rather than necessarily the amount researchers required.

Requested compute provides additional information about demand, but may also underestimate actual need if researchers limit their requests based on expected availability or if potential users do not apply.

The project therefore distinguishes between:

- requested compute,
- allocated compute,
- the observed gap between requested and allocated compute, and
- underlying or latent demand.

2.4 Project Objectives

The project aims to:

1. build a documented longitudinal dataset of Australian HPC activity,
2. improve comparability of compute measures across years, facilities and hardware generations,
3. distinguish CPU and GPU compute where possible,
4. examine historical demand, allocation and oversubscription,
5. analyse changes in Field of Research composition,
6. investigate potential structural drivers of HPC demand,
7. develop and compare suitable forecasting models,
8. forecast HPC demand to 2030 under different scenarios, and
9. estimate potential future gaps between demand and available capacity.

2.5 Analysis Framework

The analysis follows ten main steps:

1. Data collection
2. Potential variable definition
3. Data integration
4. Data standardisation
5. Data conversion
6. Initial analysis and data manipulation
7. Model development and benchmarking
8. Model evaluation
9. Model selection

10. Scenario forecasting and capacity-gap analysis

Chapter 3

Data Collection

3.1 Purpose

The first step of the project was to collect historical data on Australian HPC demand and allocation, together with other variables that may help explain changes in research compute demand over time.

The data collection focused on building a longitudinal dataset that could support both descriptive analysis and forecasting.

3.2 Data Sources

Data was collected from several sources, including:

- National Computational Merit Allocation Scheme (NCMAS) outcomes and supporting documentation,
- National Computational Infrastructure (NCI) annual reports and publications,
- Pawsey Supercomputing Research Centre annual reports and publications,
- Australian Research Council (ARC) funding data,
- National Health and Medical Research Council (NHMRC) funding data,
- Medical Research Future Fund (MRFF) data,
- research workforce and research activity data, and
- Field of Research (FoR) information.

International HPC data will also be considered where suitable data is available, including information from EuroHPC, NSF ACCESS and ARCHER2.

3.3 HPC Demand and Allocation Data

The main HPC dataset currently covers the period from 2013/14 to 2025/26.

The collected information includes, where available:

- requested compute,
- allocated compute,
- facility,
- HPC system,
- resource type,
- original compute unit,
- Field of Research,
- institution,
- source, and
- reporting period.

The level of detail varies across years because historical reporting practices and available public information are not consistent throughout the full period.

3.4 Field of Research Data

Field of Research information was collected to examine how the composition of HPC use has changed over time.

Where possible, historical FoR classifications are mapped to a consistent classification to support comparison across years.

The available FoR data primarily describes the composition of allocated compute. It should therefore not automatically be interpreted as the composition of underlying research demand.

3.5 Potential Demand Drivers

Additional datasets are being collected to investigate structural drivers of HPC demand.

These include research funding, workforce, research activity and other indicators that may be associated with changes in research compute requirements.

The suitability of these variables for forecasting will be assessed later based on their historical coverage, consistency and relationship with HPC demand.

3.6 Data Provenance

The original source, reporting period, reported unit and relevant notes are retained wherever possible.

This allows subsequent transformations and assumptions to be traced back to the original data source and helps identify observations that may not be directly comparable across years.

3.7 Data Collection Limitations

The main limitations identified during data collection include:

- limited consistent data before 2013,
- differences in reporting formats across years,
- inconsistent availability of requested and allocated compute,
- incomplete separation of CPU and GPU resources in some historical data,
- changes in HPC systems and accounting methods, and
- differences in FoR classifications over time.

These issues are addressed in the subsequent data integration, standardisation and conversion stages.